

TOI-4145Ab

R-2142

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TOI4145_250924 · created 2026-08-12T00:21:53+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2460942.863 +/- 0.045 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.135 +/- 0.033
Transit depth (Rp/Rs)^2	1.81 +/- 0.89 [%]
Orbital Inclination (inc)	89.97 +/- 1.69
Airmass coefficient 1 (a1)	0.9524 +/- 0.0065
Airmass coefficient 2 (a2)	0.02 +/- 0.17
Scatter in the residuals of the lightcurve fit is	0.69 %
Best Comparison Star	#2 - [231, 217]
Optimal Aperture	2.9
Optimal Annulus	7.76
Transit Duration (day)	0.045 +/- 0.029

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.135 $\pm$ 0.033 vs 0.14195069566578392 (deviation 0.13 σ)
Duration fit vs published	0.045 d vs 0.0875 d (deviation 0.89 σ)
Mid-time O–C	35.3 min
Depth SNR	2.5
Residual scatter	0.69 %

Light curve

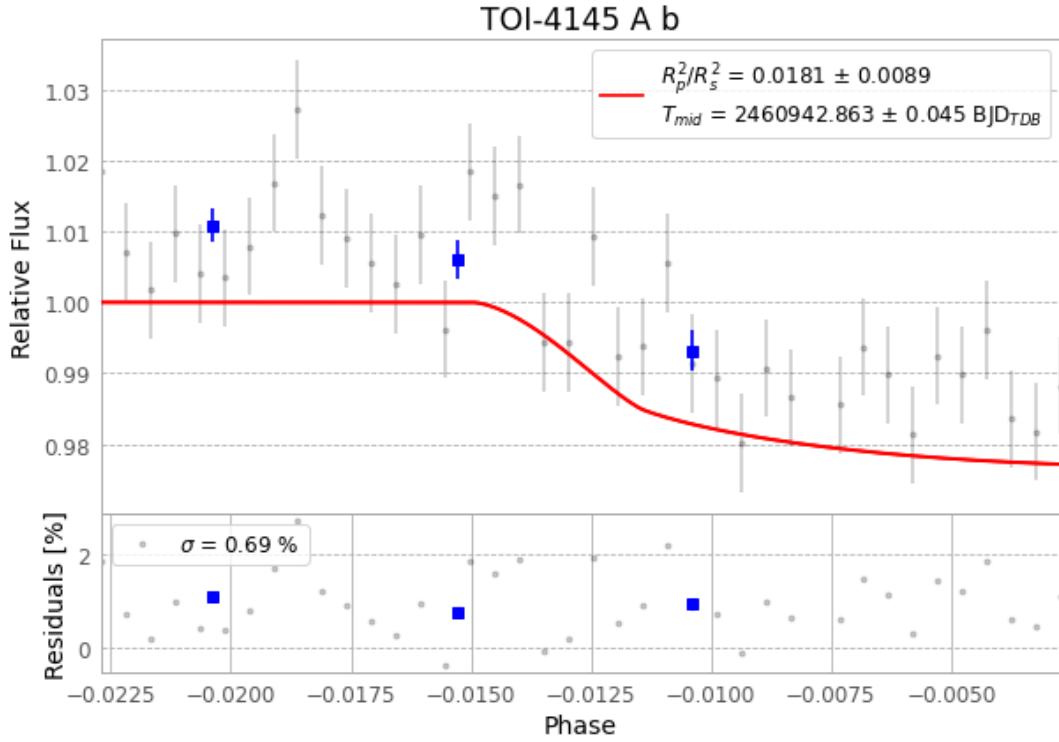


Figure 1 — FinalLightCurve_TOI-4145 A b_2025-09-23.png (SHA-256 71ae68d2999de231...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [448, 260], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 06468f7f9c45a3ac...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit fd74a91

Data provenance

40 FITS frames (26 MB), TOI4145250924062715.FITS → TOI4145250924082415.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TOI4145-250924.log (48546c8900e7...), FinalLightCurve_TOI-4145 A b_2025-09-23.png (71ae68d2999d...), FinalLightCurve_TOI-4145 A b_2025-09-23.csv (d9f8634e0242...)

Compute resources

Wall time 106.7 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-12 00:21Z.